

Alejandro de la Concha Duarte

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with a PhD from École Normale Supérieure Paris-Saclay, specializing in statistical machine learning, time-series analysis, federated learning, and non-parametric statistics. Experienced in designing and deploying machine learning algorithms for applications in finance, actuarial science, and geohazard monitoring. Recognized for strong analytical and communication skills, with a proven ability to deliver scalable, research driven solutions in both academic and industrial contexts.

SKILLS

- **Software:** Python (NumPy, SciPy, scikit-learn, TensorFlow), C++, R, SQL, CUDA, Git
- **Methods:** Time-series analysis, network analysis, kernel methods, deep learning, parametric and non-parametric statistics, graph-signal processing
- **Domains:** Risk management, actuarial analysis in life and non life insurance, geohazard monitoring.
- **Languages:** Spanish: native speaker. English: C1 (IELTS 7.5/9). French: B2 (DEL F 73/100). Italian: B2 (CELI 3: 167/200)
- **Soft skills:** Analytical thinking, communication skills, interdisciplinary collaboration, adaptability, project management

PROFESSIONAL EXPERIENCE

• SEP 2025–CURRENT (FULL TIME) **Postdoctoral Researcher**

Department of Mathematics, University of Luxembourg, Luxembourg

- Research on privacy-preserving machine learning and federated learning

• FEB 2020–FEB 2022 (PART TIME) **Research Data Scientist**

Financial Network Analytics (FNA), London, United Kingdom

- Developed machine learning algorithms for market and systemic risk management, anomaly detection, NLP and network analysis
- Delivered solutions aligned with emerging research and clients' specific needs
- Collaborated closely with engineering team to deploy models in production

• APR 2019–SEP 2019 (FULL TIME) **Master's Internship — Statistical Learning Applied to Time-Series on Graphs**

CMLA, École Normale Supérieure Paris-Saclay, Cachan, France

Project: *Segmentation of multivariate signals observed over graphs*

- Conducted a literature review on change-point detection and graph-signal processing
- Implemented and evaluated recent algorithms in Python

• APR 2018–JUNE 2018 (FULL TIME) **Data Science Intern (M1) — Statistical Learning Applied to Financial Networks and Risk Management**

Financial Network Analytics (FNA), London, United Kingdom

Project: *Tools from Network Theory Applied to Systemic Risk*

- Developed systemic risk indicators, network filtering algorithms, and simulations to model contagion and dynamic correlations in financial networks
- Conducted data-driven analyses of macroeconomic events (e.g., US vs China trade war, Greek debt crisis) and their systemic impact
- Designed interactive dashboards and visualizations; contributed to white paper writing and stakeholder reporting

• APR 2016 – SEP 2017 (FULL TIME) **Risk Analyst**

Central Bank of Mexico (BANXICO). Risk Management Directorate, Mexico City, Mexico

- Managed financial risk for Mexico's international reserves portfolio (USD 150+ billion positions) and designed credit risk indicators for the central bank's counterparties
- Conducted research on systemic risk in the Mexican banking sector and performed actuarial assessments for the bank's private pension scheme
- Presented methodologies and results at international conferences, contributing to the visibility of internal research initiatives

• OCT 2015 – MAR 2016 (FULL TIME) **Actuarial Intern (Undergraduate)**

Central Bank of Mexico (BANXICO). Risk Management Directorate, Mexico City, Mexico

- Developed mathematical models to estimate liability limits for Banco de México's insurance contracts for the 2017 fiscal year

• JUN 2015 – SEP 2015 (FULL TIME) **Research Intern — Actuarial Science**

University of Toronto, Toronto, Canada

Project: *Insurance loss modelling with Erlang-based mixture models*

- Tasks: Reviewed the literature on copula models, mixture modes and GAM applied to non life insurance

EDUCATION

- SEP 2019 – MAY 2025 **PhD in Applied Mathematics (Artificial Intelligence)**

Centre Borelli, École Normale Supérieure Paris-Saclay, Gif-sur-Yvette, France

Thesis: *Graph-Based Machine Learning for Detection Tasks in Complex Systems*.

- SEP 2018 – SEP 2019 **MSc Year 2 in Mathematics, Vision and Learning (MVA)**

École Normale Supérieure Paris-Saclay, Cachan, France

FINAL GPA: 16/20 (HIGH HONORS).

- AUG 2017 – AUG 2018 **MSc Year 1 in Mathematics and Applications**

Université d'Évry Val d'Essonne, Évry, France

FINAL GPA: 17/20 (HIGH HONORS). RANKED 1ST IN CLASS.

- AUG 2011 – AUG 2015 **BSc in Actuarial Science**

Universidad Nacional Autónoma de México (UNAM), Mexico City, Mexico

FINAL GPA: 9.97/10. RANKED 1ST IN CLASS.

- SEP 2014 – JAN 2015 **Semester 1 of BSc Year 3 in Mathematics (academic exchange)**

Université de Rouen, Rouen, France

FINAL AVERAGE: 17/20. RANKED 1ST IN CLASS.

AWARDS & HONORS

- 2017 **Gabino Barreda Medal**

UNAM Awarded for achieving the highest GPA among 410 students.

- 2014 **Second Place — National Young Statisticians Competition, Mexican Statistical Association**

Project: *Statistical analysis of characteristics associated with students who are regular readers at Universidad Veracruzana*.

- Scholarships: DIM Math'Innov, Sophie Germain Academic Excellence Scholarship, Telmex Foundation Academic Excellence Scholarship, Mitacs Globalink

PUBLICATIONS (SELECTED)

- **de la Concha Duarte, Alejandro**, Vayatis, N., Kalogeratos, A. *Collaborative likelihood-ratio estimation over graphs*, JMLR 2025.
- **de la Concha Duarte, Alejandro**, Vayatis, N., Kalogeratos, A. *Online Non-parametric Change-point Detection via Graph-based Likelihood-Ratio Estimation*, EUSIPCO 2025.
- **de la Concha Duarte, Alejandro**, Vayatis, N., Kalogeratos, A. *Collaborative non-parametric two sample testing*, AISTATS 2025
- **de la Concha Duarte, Alejandro**, Vayatis, N., Kalogeratos, A. *Online non-parametric likelihood-ratio estimation by Pearson divergence functional minimization*, AISTATS 2024
- **de la Concha Duarte, Alejandro**, Vayatis, N., Kalogeratos, A. *Offline detection of change-points in the mean for stationary graph-signals*, AISTATS 2021

OPEN SOURCE PROJECTS

Projects available in my Github profile:

- **Collaborative Likelihood-Ratio Estimation over Graphs**. Implements a framework for density and likelihood-ratio estimation on graphs, including a derived multiple hypothesis testing method (Collaborative Two Sample Testing).
- **OLRE**. Online non-parametric likelihood-ratio estimator based on Pearson divergence functional minimization.
- **VSGS**. Offline detection of change-points in the mean for stationary graph-signals using a model selection approach.
- **ADAPT (contributor)**. Python toolbox for domain adaptation algorithms.